AJATHAN

Backend Software Developer · Go & Microservices

raviajathan.2003@gmail.com | linkedin.com/in/ajathan | +91 6379857346

PROFESSIONAL SUMMARY

Backend developer with experience building and maintaining production REST APIs and microservices in Go, TypeScript, and Node.js for real-time security and fleet management platforms. Comfortable across the stack from database design (MSSQL, PostgreSQL, MySQL, MongoDB, Redis) to CI/CD and containerized deployment. Hands-on experience with IoT hardware integration, geofencing, encrypted authorization workflows, and low-latency systems that sync with live device and telematics data streams.

WORK EXPERIENCE

Junior Software Developer — *Sterna Security Devices* *Feb 2024 – Present*

- Designed and maintained multiple production REST APIs in Go (Gin/GORM) for a real-time security platform, improving data-access performance and reducing average API latency.
- Built scalable, high-availability microservices using Go, NestJS, and TypeScript; enforced clean architecture patterns and conducted regular peer code reviews.
- Integrated Redis caching to reduce database load on frequently accessed endpoints, improving throughput under peak traffic.
- Wrote unit and integration tests using Go's testing package and Testify, maintaining consistent coverage across core service modules.
- Containerized services with Docker and managed CI/CD pipelines and version-control workflows using Git with feature-branch strategies.
- Collaborated with the security hardware team to integrate backend APIs with embedded device data streams, ensuring low-latency real-time updates.

Web Developer Intern — *Futurik Technologies* *Jun 2023 – Nov 2023*

- Built and maintained responsive web interfaces using HTML, CSS, and Tailwind CSS, delivering UI components for client-facing projects.
- Assisted in backend feature development using Node.js, gaining hands-on exposure to REST API design and database querying with MySQL.

TECHNICAL SKILLS

Languages	Go, TypeScript, JavaScript, SQL
Frameworks & Libraries	Gin (Gonic), NestJS, Node.js, Express
Databases & ORMs	MSSQL, PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, GORM
API & Architecture	REST, OpenAPI / Swagger, Microservices, RBAC, Request Validation
IoT & Telematics	Smart Lock Integration (Duco 2, CLIQ), Geofencing, VTS, AES Encryption, HOTP/OTP, RTC Time-Sync
DevOps & Tools	Docker, Git, GitHub, Postman, Azure Data Studio, Linux, VS Code
Testing	Go testing package, Testify
Cloud (Familiar)	AWS (EC2, S3 basics)
Web	HTML5, CSS3, Tailwind CSS

PROJECTS

UNIGEM3 & UNIGEM4 — Fleet Management Systems

- Built microservices in Go (Gin) and Node.js (NestJS) to handle real-time IoT lock events, trip tracking, and user authorizations for a fleet management platform.
- Wrote advanced SQL queries in MSSQL to run fast, complex authorization checks across large datasets like key audit trails and truck location status.
- Connected hardware smart locks (Duco 2, CLIQ cylinders) to backend services, including encrypted data handling, time synchronization, and multi-factor OTP verification.

- Built geofencing logic that tracks when a vehicle enters or exits a terminal or route, and uses that to automatically authorize lock operations.
- Tracked down and fixed data mismatches between services and database constraints to keep the system reliable.

Twitter Sentiment Analysis

- Built a machine learning pipeline to classify tweet sentiment (positive/negative/neutral) using Python, scikit-learn, and NLP preprocessing techniques.

Flight Ticket Booking Site

- Developed a full-stack flight booking web app with search, seat selection, and booking management; backend built with Node.js and MySQL.

EDUCATION

Dr. Mahalingam College of Engineering and Technology *2020 – 2024*
B.E. Computer Science and Engineering

ACHIEVEMENTS & CERTIFICATIONS

- Presented a research paper on AI in Healthcare at a national-level academic conference.
- Runner-up, DESIGN CHAMPIONSHIP 2018 (Game Design) — conducted by Unity & NASSCOM.
- Completed a masterclass on SQL vs NoSQL: database design trade-offs and query optimization.

SOFT SKILLS

Strong communicator • Collaborative team player • Analytical problem-solver • Reliable with deadlines